Network Security Laboratory

Experiment 4: Configuring and Testing Access Control Lists (ACLs)

Department	Cybersecurity Engineering Technologies
Stage	Fourth Year
Software	Cisco Packet Tracer
Goal	Block ICMP ping from PC0 to PC1 while allowing other IP traffic

1. Objective

Configure an extended ACL on router R1 to prevent PC0 (192.168.1.10) from sending ICMP ping packets to PC1 (192.168.2.10), while permitting all other IP traffic.

2. Required Devices

- 1 × Cisco 2911 Router
- 2 × Cisco 2960 Switches
- 2 × PCs

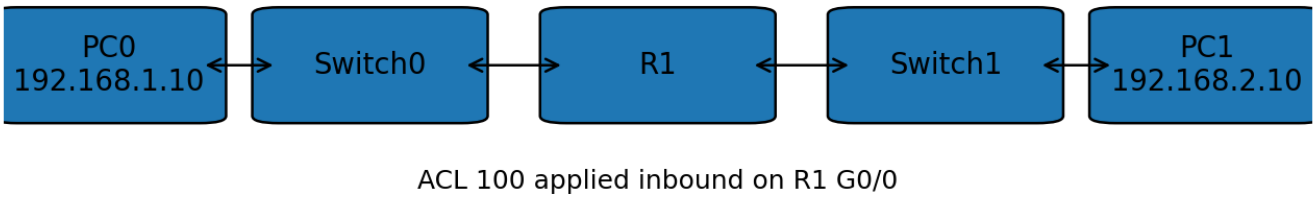


Figure 1. Lab topology and ACL placement

3. IP Addressing

Device	Interface	IP Address	Default Gateway
PC0	Fa0	192.168.1.10/24	192.168.1.1
R1	G0/0	192.168.1.1/24	—
R1	G0/1	192.168.2.1/24	—
PC1	Fa0	192.168.2.10/24	192.168.2.1

4. Build and Test the Network

- Connect PC0 → Switch0 → R1 G0/0.
- Connect R1 G0/1 → Switch1 → PC1.
- Configure PC0 and PC1 using the addressing table.
- Use the automatic cable option in Packet Tracer if needed.

5. Router Configuration

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
end
```

6. Test Before Applying the ACL

From PC0, open Command Prompt and run:

```
ping 192.168.2.10
```

Expected result: the ping should succeed because no ACL is blocking the traffic yet.

7. Configure ACL 100

```
enable
configure terminal
access-list 100 deny icmp host 192.168.1.10 host 192.168.2.10
access-list 100 permit ip any any
interface gigabitEthernet 0/0
 ip access-group 100 in
end
write memory
```

8. How the ACL Works

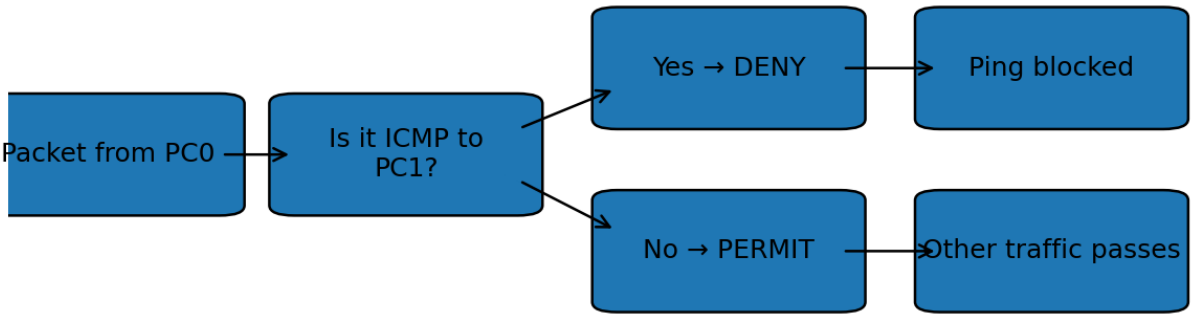


Figure 2. Simplified ACL decision flow

Command	Purpose
deny icmp host 192.168.1.10 host 192.168.2.10	Blocks ping from PC0 to PC1
permit ip any any	Allows all other IP traffic
ip access-group 100 in	Applies ACL 100 to packets entering R1 from PC0 side

9. Testing After Applying the ACL

On PC0:

ping 192.168.2.10

Expected result:

Request timed out.

This confirms that ACL 100 is blocking ICMP traffic from PC0 to PC1.

10. Verify the ACL

show access-lists

Expected ACL entries:

Extended IP access list 100

deny icmp host 192.168.1.10 host 192.168.2.10

permit ip any any

To verify where the ACL is applied:

show ip interface gigabitEthernet 0/0

11. Remove the ACL for Testing

enable

configure terminal

interface gigabitEthernet 0/0
 no ip access-group 100 in
end
Repeat the ping. It should work again.

12. Student Task

- Create the topology.
- Configure all IP addresses.
- Verify ping works before applying the ACL.
- Configure ACL 100.
- Apply ACL 100 inbound on R1 G0/0.
- Verify that ping from PC0 to PC1 fails.
- Display the ACL using show access-lists.
- Remove the ACL and verify ping works again.
- Save the Packet Tracer file as Week4_ACL_StudentName.pkt.

13. Required Screenshots

- Complete topology.
- Successful ping before the ACL.
- ACL 100 configuration.
- Failed ping after applying the ACL.
- Output of show access-lists.

14. Quick Troubleshooting

Problem	Check
Ping fails before ACL	IP addresses, gateways, router interfaces, cables
Ping still works after ACL	ACL direction and interface: G0/0 inbound
All traffic is blocked	Ensure permit ip any any exists

15. Lab Questions

1. What is the purpose of an ACL?

It controls which network traffic is permitted or denied.

2. What protocol is blocked in this experiment?

ICMP.

3. What command applies ACL 100 to the interface?

ip access-group 100 in

4. Why do we use permit ip any any?

To allow all other IP traffic.

5. What is the expected ping result after applying the ACL?

Request timed out.

Expected Result

Before ACL: PC0 can ping PC1. After ACL 100 is applied inbound on R1 G0/0, the ping is blocked while other IP traffic remains permitted. Removing the ACL restores the ping.

```
enable
configure terminal
hostname R1
interface gigabitEthernet 0/0
 ip address 192.168.1.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 192.168.2.1 255.255.255.0
 no shutdown
exit
access-list 100 deny icmp host 192.168.1.10 host 192.168.2.10
access-list 100 permit ip any any
interface gigabitEthernet 0/0
 ip access-group 100 in
end
write memory
show access-lists
```